

Artificial Intelligence Homework 1

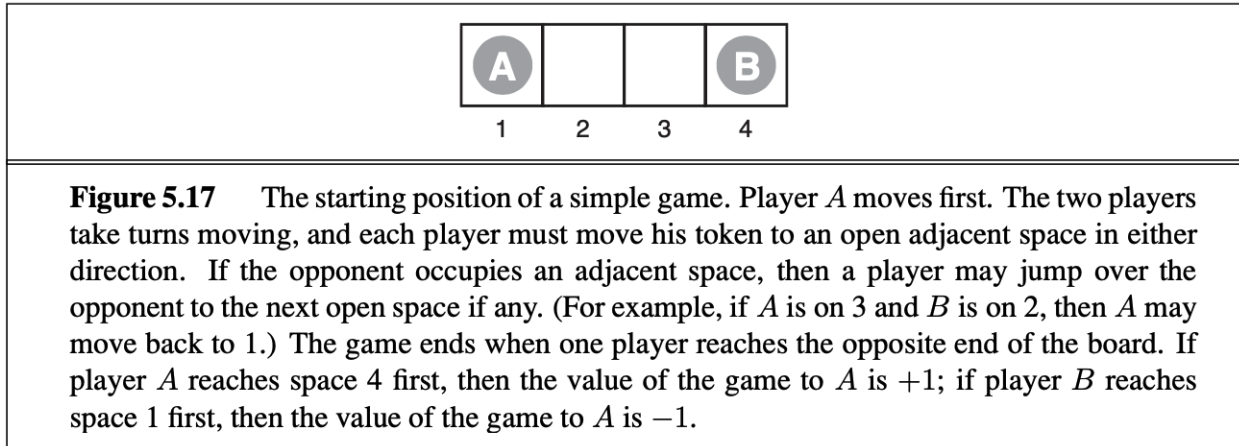
March 16, 2026

Q1: Give a complete problem formulation for each of the following. Choose a formulation that is precise enough to be implemented.

- a. Using only four colors, you have to color a planar map in such a way that no two adjacent regions have the same color.
- b. A 3-foot-tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, movable, climbable 3-foot-high crates.
- c. You have a program that outputs the message "illegal input record" when fed a certain file of input records. You know that processing of each record is independent of the other records. You want to discover what record is illegal.
- d. You have three jugs, measuring 12 gallons, 8 gallons, and 3 gallons, and a water faucet. You can fill the jugs up or empty them out from one to another or onto the ground. You need to measure out exactly one gallon.

Q2: Consider a state space where the start state is number 1 and each state k has two successors: numbers $2k$ and $2k + 1$.

- a. Draw the portion of the state space for states 1 to 15.
- b. Suppose the goal state is 11. List the order in which nodes will be visited for breadth-first search, depth-limited search with limit 3, and iterative deepening search.
- c. How well would bidirectional search work on this problem? What is the branching factor in each direction of the bidirectional search?
- d. Does the answer to (c) suggest a reformulation of the problem that would allow you to solve the problem of getting from state 1 to a given goal state with almost no search?
- e. Call the action going from k to $2k$ Left, and the action going to $2k + 1$ Right. Can you find an algorithm that outputs the solution to this problem without any search at all?



Q3: Consider the two-player game described in Figure 5.17.

- a. Draw the complete game tree, using the following conventions:
 - Write each state as (s_A, s_B) , where s_A and s_B denote the token locations.
 - Put each terminal state in a square box and write its game value in a circle.
 - Put loop states (states that already appear on the path to the root) in double square boxes. Since their value is unclear, annotate each with a "?" in a circle.
- b. Now mark each node with its backed-up minimax value (also in a circle). Explain how you handled the "?" values and why.
- c. Explain why the standard minimax algorithm would fail on this game tree and briefly sketch how you might fix it, drawing on your answer to (b). Does your modified algorithm give optimal decisions for all games with loops?
- d. This 4-square game can be generalized to n squares for any $n > 2$. Prove that *A* wins if n is even and loses if n is odd.

Q4: Consider the problem of solving two 8-puzzles.

- a. How large is the reachable state space? Give an exact numerical expression.
- b. Suppose we make the problem adversarial as follows: the two players take turns moving; a coin is flipped to determine the puzzle on which to make a move in that turn; and the winner is the first to solve one puzzle. Which algorithm can be used to choose a move in this setting?
- c. Give an informal proof that someone will eventually win if both play perfectly.